# Two-pan weight kits

## The mathematics in one view

The target stays on the left pan. Each positive whole-number weight is available once and can be placed on the left, on the right, or off the balance. A target t is possible exactly when $t = \Sigma\, \varepsilon_i w_i$ with $\varepsilon_i$ in {-1,0,1}; a positive sign means opposite the target, a negative sign means beside it.

**Balanced ternary.** The kit $1,3,9,\ldots,3^{m-1}$ balances every integer target from 1 through $(3^m-1)/2$, in exactly one placement with the target fixed. For two, three, four and five weights, the endpoints are 4,13,40,121.

**Optimality.** No m-weight kit can balance more than $(3^m-1)/2$ distinct positive targets, even if gaps are allowed. There are $3^m$ placement choices; all-off gives zero, and reversing the two pans pairs each positive difference with a negative one. Collisions and additional zero differences only reduce the count. This counts placements in one exact balance, not outcomes of successive weighings.

**Why powers of three fit.** If the smaller weights have total S and represent every integer from -S through S uniquely, adding 2S+1 gives three disjoint adjacent intervals and unique coverage from -(3S+1) through 3S+1. A heavier new weight leaves a gap; a lighter one cannot reach as far.

## What each group is doing

**K-1:** Move weights, compare small kits, and discover that adding beside the target can help. Count totals to 6 initially; the last problem reaches 13 with adult support. Negative numbers and equations are unnecessary.

**Grades 2-3:** Compare overlap, gap and exact fit for third weights 8,9,10; formulate the next-weight rule. Addition to 13, then a way to reason about intervals to 40, is helpful. The placement-count bound can be a shared adult-led explanation.

**Grades 4-5:** Explain uniqueness and the universal counting bound, distinguish many targets from an unbroken run, and extend to five weights. Powers can be counted as repeated groups of three; algebra is optional.

## Materials and preparation

Weights that check themselves: linking-cube sticks (or Cuisenaire rods, as in Week 29) of 1, 2, 3, 4, 7, 8, 9 and 10 units (44 cubes per kit), target sticks in another color, and a two-pan mat on which each pan's sticks lie end to end in one row. Equal row lengths balance, so children can see a balance rather than add. Snap new sticks when a child chooses 5, 6, 7, 11 or another weight. Keep paper cards for 27, 81 and big targets, plus blank cards to write on. Five pair-kits, pencils; give only the current kit, since two 1-sticks would break the rules. A real equal-arm balance is optional. This kit has not been rehearsed: check that the sticks lie flat and the rows fit the mat.

Print the chosen entry page for each child and later pages on demand. Allow about 15 minutes to snap the weight sticks for five kits (about 220 cubes) and set out cubes of another color for targets. All current adaptations are unpiloted.

## Launch and a flexible hour

**Launch, about 5 minutes after movement:** Let pairs move sticks freely, then show the printed example: target 5 with weights 2 and 7. Put 2 beside the target and 7 opposite, so 5+2=7, and lay the rows side by side to show they match. Leave an unused weight off. Ask a child to explain why the totals agree.

**About 35-40 minutes at tables.** First 15-20 minutes: K-1 P1-3; 2-3 P1-3; 4-5 P1-2. Then pairs pose a target for the other pair at their table (at the 4-5 table two build and the third checks, rotating). Continue kit comparisons or move to duplicate balances, uniqueness and the counting bound. In the last minutes, show one useful balance and one reason a kit must fail. Stop at small examples if extending the kit would replace understanding with arithmetic.

## Small-kit key: shared across levels

| Kit | All positive targets it balances | Reason |
|---|---|---|
| 1,2 | 1,2,3 | Single weights, sum 3, difference 1 |
| 1,3 | 1,2,3,4 | 2+1=3; also singles and sum 4 |
| 1,4 | 1,3,4,5 | 3+1=4; misses 2 |
| 1,3,8 | 1 through 12 | Old differences -4...4, plus 8: 4...12 |
| 1,3,9 | 1 through 13 | Old differences -4...4, plus 9: 5...13 |
| 1,3,10 | 1...4 and 6...14 | New positive interval starts at 10-4=6 |

## K-1 answer key

**P1.** 1=1, 2+1=3, 3=3, 4=1+3. The target is the first number on each left side; each listed weight is used once.

**P2.** Exactly 1,2,3 work with 1 and 2; 4 and 5 exceed their total weight 3.

**P3.** A tie in the *number* of targets: each kit reaches four of 1-6. The 1/4 kit reaches 1,3,4,5; the 1/3 kit reaches 1,2,3,4. Only the latter has no gap in its run.

**P4.** Only {1,3}. All six kits give: {1,2} → {1,2,3}; {1,3} → {1,2,3,4}; {1,4} → {1,3,4,5}; {2,3} → {1,2,3,5}; {2,4} → {2,4,6}; {3,4} → {1,3,4,7}. Every other kit misses a requested target.

**P5.** Choose 9. The 8-kit cannot reach 13 because its total is 12; the 10-kit cannot reach 5 because even putting both small weights beside 5 leaves 5+4<10. The 9-kit constructions are on the next page.

## Grades 2-3 answer key

**P1.** Use the small-kit table on page 2. A larger maximum is not necessarily a longer gap-free run.

**P2.** Choose 9. With a new weight w, reaching 13 requires w+4≥13, hence w≥9. Reaching 5 requires w-4≤5, hence w≤9; a small kit without w reaches at most 4. Thus 9 is the only choice.

| Target | Balanced pans | Target | Balanced pans |
|---|---|---|---|
| 1 | 1 = 1 | 8 | 8+1 = 9 |
| 2 | 2+1 = 3 | 9 | 9 = 9 |
| 3 | 3 = 3 | 10 | 10 = 9+1 |
| 4 | 4 = 3+1 | 11 | 11+1 = 9+3 |
| 5 | 5+3+1 = 9 | 12 | 12 = 9+3 |
| 6 | 6+3 = 9 | 13 | 13 = 9+3+1 |
| 7 | 7+3 = 9+1 | | |

In this table the first number on the left is the target, not another weight. Unmentioned weights stay off.

**P3.** First missed positive targets: 1/3/8 → 13; 1/3/9 → 14; 1/3/10 → 5. The complete ranges on page 2 establish that these are the *first* misses.

**P4.** With 1,3,8 there are exactly two balances of 4: 4=3+1 (8 off), and 4+3+1=8. With 1,3,9 there is exactly one: 4=3+1 (9 off). If the big weight is opposite the target, the small weights must supply the remaining difference; if it is beside the target, the opposite small weights cannot outweigh it. These cases exhaust the placements.

**P5.** No. Three weights have 27 placements, at most 13 distinct positive differences after zero and sign reversal are accounted for. Fourteen different positive targets require at least 29 placements.

**P6.** Add 27. The old kit gives every signed difference from -13 through 13. Put 27 opposite and use the old kit to adjust by any of these differences: this covers 14-40. Together with 1-13 it covers 1-40. A fourth weight below 27 has total below 40; above 27 it misses 14.

## Hints, only when needed

For a stuck balance, ask whether a weight may sit beside the target. For comparing new weights, find the smallest target using the new weight and the largest using it. For “all ways,” split into big weight off / opposite / beside, rather than move cards randomly.

## Grades 4-5 answer key and proofs

**P1.** Only {1,3}. For 0<a<b the possible positive targets are among a, b, b-a, a+b. To cover 1,2,3,4, these four values must be distinct and equal that set. Their maximum a+b must therefore be 4; the only distinct positive integer pair with that sum is 1,3.

**P2.** Add 9 to cover 1-13. If w<9, total w+4<13. If w>9, 5 is impossible: w opposite gives at least w-4>5; w beside cannot produce a positive target, and omitting w reaches at most 4.

**P3.** No duplicate placements exist for any positive target 1-13. Suppose two placements gave the same difference. Subtract them and take the largest weight where they differ. Its contribution has magnitude at least that weight; all smaller changes together have magnitude at most twice their sum. For 9 this is 2(1+3)=8<9; for 3 it is 2<3; for 1 it is 0<1. Cancellation is impossible.

**P4.** No, by the 27-placement / 13-positive-target bound (page 3). It applies to any three weights, regardless of their magnitudes or whether some placements give zero.

**P5.** Counts depend on the chosen kit, but the maximum is 13. For example, 1,2,3 reaches only 1-6. The kit 1,3,9 attains 13 with no gaps. The kit 1,4,16 also attains 13 but reaches {1,3,4,5,11,12,13,15,16,17,19,20,21}. Why none can do better: the 27 placements include all-off, which gives zero, and the rest pair into opposites by swapping the pans, so at most 13 are positive. A kit cannot exceed 13 merely by allowing gaps.

**P6.** Add 27: the signed intervals are [-40,-14], [-13,13], [14,40]. They are disjoint and adjacent, with a unique old-kit placement inside each interval, so the new kit has neither gaps nor duplicates. Alternatively 27>2(1+3+9)=26 proves uniqueness. The 81 placements bound the positive run by 40, so it is optimal.

**P7.** The maximum is $(3^5-1)/2=121$, achieved by 1,3,9,27,81. The old signed interval is [-40,40]; adding 81 gives [-121,-41], [-40,40], [41,121], each uniquely represented. The 243 placement choices permit at most 121 positive targets.

## Optional extension

If the group owns the argument, ask what changes when weights may only go opposite the target. Each has two choices; 1,2,4 cover 1-7 uniquely. This contrasts two versus three placement choices without introducing a different weighing puzzle.

## Primary source

Ethan D. Bolker, Samuel A. Feuer and Catalin Zara, “Balance weighing - variations on a theme,” *Mathematics Magazine* 94(5), 2021, author manuscript pp. 1-2, Theorem 1. It reports fourth graders exploring the two-pan sequence and states balanced-ternary uniqueness. Current finite cases and the placement-count proof were checked independently. Read the author-hosted paper.

# Starting routes and return visits

Planning update of 4 October 2026. The timing menus are examples; a theme can continue over several meetings. Keep the full Grades 4-5 investigations and choose by readiness.

### Use and return choices

Keep one adult at each table; paper cards represent amounts and the adult may check small sums. On another visit, K-1 can choose among two-weight kits (P4) or add a third weight (P5). Grades 2-3 can compare repeated balances (P4) before the counting bound and fourth-weight kit (P5-6). Grades 4-5 can choose uniqueness (P3), a universal placement bound (P4-5), or the four-/five-weight construction (P6-7). Reuse the same pan convention and deal only the weights in the current kit.